

* Embedding space $\mathcal{M} \equiv \{x^\mu, \chi^A\}$, $\mu=0, \dots, D-1$
 $\text{Span}\{\chi^A\} = \mathbb{M}^D$

χ^A : Grassmann odd, 2D Majorana spinors
 $\hookrightarrow$ Not a spinor of $\mathcal{M}$ but rather a D-dim vector over $\mathcal{M}$

χ_A^A : $\text{Spin}(2)$ of Σ
D of $\text{SO}(D)$ in $\mathcal{M}$

• Theory possesses (1,1) susy on Σ , meaning we have χ_A^μ and $\chi_{\dot{A}}^\mu$

* Off-shell we have Aux fields N^μ

$$S = \int d^2 \Sigma \left(-\frac{1}{2} \partial_\alpha x^\mu \partial^\alpha x^\nu - \frac{i}{2} \bar{\chi}^\mu \not{\partial} \chi^\nu + \frac{1}{2} N^\mu N^\nu \right) \mathcal{R}_{\mu\nu}$$

Supermultiplet $\{x^\mu, \chi_A^\mu, N^\mu\}$

α : Σ index (vector) over $\mathbb{M}^2$

μ : $\mathcal{M}$ index

A : Σ index (spinor)

a : Σ index (vector) over General M^2

• Above admits global susy, i.e.

$$\delta_\epsilon = \epsilon^\alpha Q_\alpha + \bar{\epsilon}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}}$$

s.t. $\epsilon^\alpha \neq \epsilon^\alpha(x)$

* Using the Noether method - couple the theory to sugra $\{e_\alpha^a, \psi_{\alpha A}\}$

$\rightarrow$ Superstring action:

$$S = \int d^2 \Sigma \left(-\frac{1}{2} \partial_\alpha x^\mu \partial^\alpha x^\nu - \frac{i}{2} \bar{\chi}^\mu \not{\partial} \chi^\nu + \frac{1}{2} N^\mu N^\nu \right) \mathcal{R}_{\mu\nu}$$

$$S = 2\pi\alpha' \int d^2\zeta \left\{ -\frac{1}{2} e_a \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} - \frac{1}{2} \bar{\psi} \gamma^\alpha \partial_\alpha \psi - \frac{\kappa^2}{16} \bar{\psi} \gamma^\alpha \gamma^\beta \psi \partial_\alpha X^\mu \partial_\beta X^\nu \right\}$$

$$= \int d^2\zeta \mathcal{L}$$

$$e = \det(e_a^\alpha) \quad \mathcal{L} = \gamma^a e_a^\alpha \partial_\alpha$$

Symmetries of the theory:

① 2D reparametrization invariance

$$\begin{aligned} \delta e_a^\alpha &= f^\beta \partial_\beta e_a^\alpha + \partial_\alpha f^\beta e_{\beta a}^\alpha \\ \delta \psi_{\alpha A} &= f^\beta \partial_\beta \psi_{\alpha A} + \partial_\alpha f^\beta \psi_{\beta A} \\ \delta X^\mu &= f^\beta \partial_\beta X^\mu \\ \delta \chi_A^\mu &= f^\beta \partial_\beta \chi_A^\mu \\ \delta N^\mu &= f^\beta \partial_\beta N^\mu \end{aligned}$$

f^a : local, world-sheet dependent parameter

Proof:

$$\delta A = \int d^2\zeta \left\{ \delta e \right\} + \int d^2\zeta e \left\{ \delta \right\}$$

① $\delta e = \delta \det(e)$, $\det(e) = e^{\text{tr}(\ln(e))}$

$$\begin{aligned} \delta \det(e) &= e \text{tr}(\delta \ln(e)) \\ &= e (e^{-1})^\alpha_a \delta e_a^\alpha \\ &= e e_a^\alpha \delta e_a^\alpha \end{aligned}$$

$$= \int d^2\zeta e \left\{ e_a^\alpha \delta e_a^\alpha \right\}$$

② ② $-\frac{1}{2} \delta (e_a^\alpha \partial_\alpha X^\mu e^{a\beta} \partial_\beta X^\nu)$

$$\begin{aligned} &= -\frac{1}{2} \left[\delta e_a^\alpha \partial_\alpha X^\mu e^{a\beta} \partial_\beta X^\nu \right. \\ &\quad \left. + e_a^\alpha \partial_\alpha (\delta X^\mu) e^{a\beta} \partial_\beta X^\nu \right. \\ &\quad \left. + e_a^\alpha \partial_\alpha X^\mu (\delta e^{a\beta}) \partial_\beta X^\nu \right] \end{aligned}$$

$$\begin{aligned}
 & + e_a^\alpha \partial_\alpha f^B \partial_B x^m e^{a\beta} \partial_\beta x^N + e_a^\alpha f^B \partial_\alpha \partial_B x^m e^{a\gamma} \partial_\gamma x^N \\
 & + e_a^\alpha \partial_\alpha x^m f^\gamma \partial_\gamma e^{a\beta} \partial_\beta x^N + e_a^\alpha \partial_\alpha x^m \partial^\beta f^\gamma e_\gamma^a \partial_\beta x^N \\
 & + e_a^\alpha \partial_\alpha x^m e^{a\beta} (\partial_\beta f^\gamma) \partial_\gamma x^N + e_a^\alpha \partial_\alpha x^m e^{a\beta} f^\gamma \partial_\beta \partial_\gamma x^N]
 \end{aligned}$$

$$\begin{aligned}
 = & -\frac{1}{2} \left[-\partial_\alpha x^m \partial_\beta x^N f^\gamma (\partial_\gamma e_\beta^a) - (\partial_\beta f^\alpha) \partial_\alpha x^m \partial^\beta x^N \right. \\
 & \quad \underline{\partial_\alpha f^B \partial_\beta x^m \partial^a x^N} + \underline{f^B \partial_\alpha \partial_\beta x^m \partial^a x^N} \\
 & \quad \partial_\alpha x^m f^\gamma (\partial_\gamma e_\beta^a) \partial^\beta x^N + \partial_\alpha x^m \partial^\beta f^\gamma e_\gamma^a \partial_\beta x^N \\
 & \quad \left. + \underline{\partial_\alpha x^m \partial^a f^\gamma \partial_\gamma x^N} + \underline{\partial_\alpha x^m f^\gamma \partial^a \partial_\gamma x^N} \right] \mathcal{Q}_m
 \end{aligned}$$

$$\bullet \partial_\alpha x^m \partial^a f^\gamma \partial_\gamma x^N + \partial^a f^\gamma \partial_\gamma x^N \partial_\alpha x^m$$

$$= 2 \partial_\alpha x^m \partial^a f^\gamma \partial_\gamma x^N$$

$$\bullet f^\gamma (\partial^a \partial_\gamma x^m) \partial_\alpha x^N + \partial_\alpha x^N f^\gamma \partial^a \partial_\gamma x^m$$